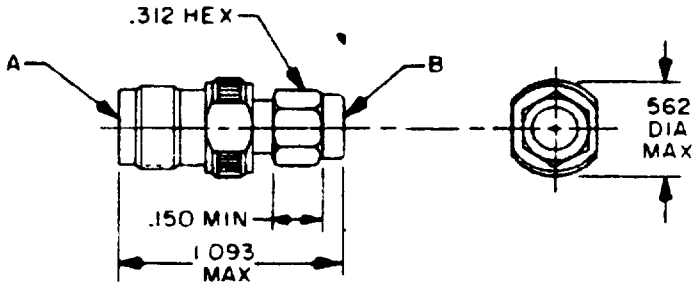
MILITARY SPECIFICATION SHEET

ADAPTERS, CONNECTOR, COAXIAL, RADIO FREQUENCY, (BETWEEN SERIES SMA (MALE) TO SERIES TNC (FEMALE)), CLASS 2, STRAIGHT PLUG

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the adapters described herein shall consist of this document and the latest issue of Specification MIL-A-55339



Reference	Series	Contact	Figure
A	TNC	Socket	2
B	SMA	Pin	3

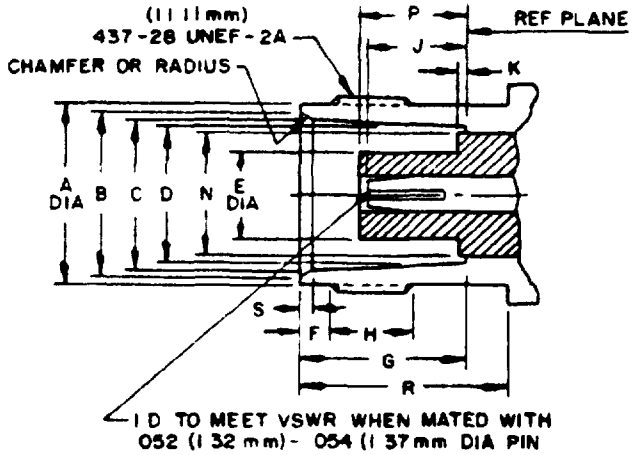
INCHES	MM
.150	3.81
.312	7.92
.562	14.27
1.093	27.76

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm
3. All unmentioned pictorial representations are for reference purposes only
4. Wrench flats to accommodate standard wrench per H-28, appendix 10

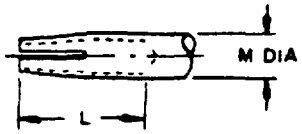
FIGURE 1. General configuration

FSC 5935

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:07Z
Check the source to verify that this is the current version before use.



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note)		
	Minimum		Maximum
A	.378	(9 60)	381 (9.68)
B	.345	(8 76)	.356 (9.04)
C	.327	(8.31)	.333 (8 46)
D	.319	(8 10)	.321 (8 15)
E			.186 (4.72)
F	.068	(1.73)	.088 (2.24)
G	.329	(8.36)	.333 (8.46)
H	.187	(4.75)	
J	186	(4 72)	.206 (5.23)
K			.006 (.15)
L	.195	(4.95)	
M	.081	(2.06)	.087 (2.21)
N			.256 (6.50)
P	188	(4 78)	.208 (5.28)
R	.415	(10 54)	
S	.015	(.38)	.030 (.76)

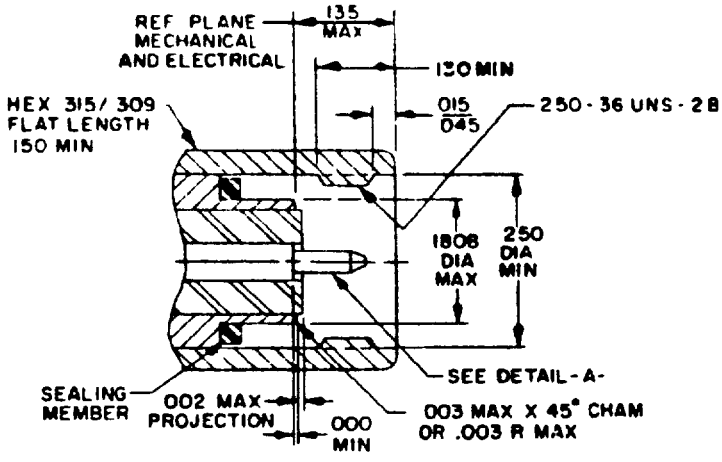


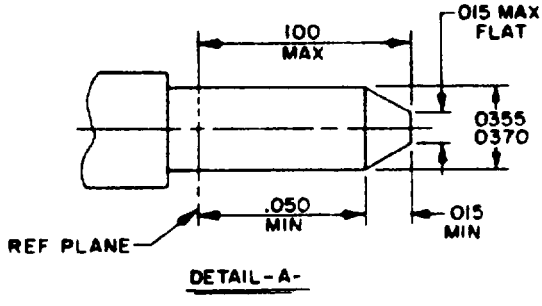
*N dimension applies to that portion (if applicable) of the dielectric which protrudes beyond the metal shoulder (or reference plane) by dimension K

NOTES
1 Metric equivalents are given for general information only and are based upon 1 00 inch = 25 4 mm
2 All undimensioned pictorial configurations are for reference purposes only

FIGURE 2 Mating dimensions for TNC contact terminations

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:07Z
Check the source to verify that this is the current version before use.





INCHES	MM
.002	05
003	08
015	38
0355	902
0370	940
045	1 14
050	1 27
.100	2 54
125	3 18
130	3 30
135	3 43
1808	4 592
250	6 35
309	7 85
315	8 00

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only and are based upon 1 00 inch = 25 4 mm
3. Three holes .016 (41 mm) minimum diameter, equally spaced, are required for safety wiring after mating Location on coupling nut optional

FIGURE 3 Mating dimensions for SMA contact terminations

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:07Z
Check the source to verify that this is the current version before use.

ENGINEERING DATA

Nominal impedance 50 ohms
Frequency range 0 to 11 GHz.
Voltage rating 335 volts rms maximum working voltage at sea level, 85 volts rms at 70,000 feet.
Temperature range -65° to +165°C.

REQUIREMENTS

Dimensions and configurations See figures 1, 2, and 3

Center contact retention
Axial force - 6 pounds minimum.

Force to engage and disengage
Longitudinal force - Not applicable.
Torque - 2 inch-pounds maximum.

Coupling proof torque
Series SMA - 15 inch-pounds, minimum.
Series TNC - Not applicable.

Inspection conditions
Coupling torque - Series SMA - 7 to 10 inch-pounds.
Series TNC - 4 to 6 inch-pounds.

Mating characteristics
Series SMA - Not applicable.
Series TNC
See figure 2 for dimensions.
Center contact (socket)
Oversize test pin - .057 inch minimum.
Insertion depth - .125 inch minimum.
Number of insertions - 1.

Insertion force test Steel test pin diameter - .054 inch minimum.
Test pin finish - 16 microinches.
Insertion force - 2 pounds maximum.

Withdrawal force Steel test pin diameter - .052 inch maximum
Withdrawal force 2 ounces minimum.
Test pin finish 16 microinches.

Permeability Not to exceed 2.

Insulation resistance 5,000 megohms minimum.

Voltage standing wave ratio (VSWR) 1.15+.015 F(GHz) at .5 to 11 GHz

RF leakage -60 dB minimum tested at a frequency of 2 to 3 GHz.

RF insertion loss .06 $\sqrt{F(GHz)}$ dB maximum at 6 GHz.

Durability 500 cycles minimum at 12 cycles per minute maximum.

Dielectric withstanding voltage 1,000 volts rms minimum at sea level

Contact resistance In milliohms maximum.

	Initial	After environment
Center contact	4.5 1/	6.0
Outer contact	2.2	---

1/ Two center contacts must be mated to the center conductor under test therefore doubling center contact resistance

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:07Z
Check the source to verify that this is the current version before use.

Vibration, high frequency. Method 204, MIL-STD-202, test condition D, interruptions -1 µs maximum.

Shock (specified pulse) Method 213, MIL-STD-202, test condition I.

Thermal shock· Method 107, MIL-STD-202, test condition C.

Moisture resistance 200 megohms minimum.

Corona level
Voltage - 375 volts minimum.
Altitude - 70,000 feet minimum.

RF high potential withstanding voltage.
RF voltage - 1,000 volts rms minimum.
Frequency - 5 to 7.5 MHz.

Salt spray (corrosion): Method 101, MIL-STD-202, test condition B.

Coupling mechanism retention force
Series TNC - Not applicable.
Series SMA - 60 pounds minimum.

Part number M55339/40-50001 (safety wire holes (SMA))
-50101 2/ (without safety wire holes)

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians
Army - EL
Navy - EC
Air Force - 85

Review activities
Army - MI, EL, AT
Navy - SH
Air Force - 11, 99
DLA - ES

User activities
Army - AT, AR
Navy - AS, MC
Air Force - 19

Preparing activity
Army - EL

Agent
DLA - ES

(Project 5935-3029-1)

2/ For logistic purposes adapters with only the safety wire holes shall be stocked

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:07Z
Check the source to verify that this is the current version before use.

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

OFFICIAL BUSINESS
LTY FOR PRIVATE USE $300

Commander
US Army Electronics Command
ATTN: DRSEL-RD-TS
Fort Monmouth, NJ 07703

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:07Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22 R255

INSTRUCTIONS The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay or will otherwise enhance use of the document DoD contractors government activities or manufacturers vendors who are prospective suppliers of the product are invited to submit comments to the government Fold on lines on reverse side staple in corner and send to preparing activity Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements Attach any pertinent data which may be of use in improving this document If there are additional papers, attach to form and place both in an envelope addressed to preparing activity

DOCUMENT IDENTIFIER AND TITLE MIL-A-55339/40A ADAPTERS, CONNECTOR, COAXIAL, RADIO FREQUENCY,)BWTWEEN SERIES SMA(MALE)TO SERIES TNC(FEMALE),CLASS 2, STRAIGHT PLUG

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

YES NO (If "Yes" in what way?)

4 REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:07Z
Check the source to verify that this is the current version before use.